

**Universal Station^X
Customer Release
Guide**

**Release 300
UX04-493**

Universal Station^x

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Customer Release
Guide**

**Release 300
UX04-493
11/99**

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About This Publication

Purpose

The Universal Station^x Customer Release Guide provides the following:

- An overview of U^xS Release 300 differences
- Documentation reference section
- Installation and upgrade preparation instructions
- Instructions for upgrading a local U^xS from HP-UX 9.05 to HP-UX 10.20
- Instructions for upgrading additional software both locally and remotely
- Instructions for upgrading a remote U^xS from HP-UX 9.05 to HP-UX 10.20

U^xS Release 300 is compatible with TPS Network Release 433, 531 or 600, or later versions of these releases.

Revision Notes

The following list provides notes concerning the current revision of this document.

Doc ID	Rev. No.	Date	Notes
UX04494	3	11/08/99	Changes made to Revision 3 of this document have been marked with change bars in the outside margin. Overview of Changes 1. Additional file added to the list of configuration and data files to be backed up (backup number two). 2. The Install U^xS Personality on a Local Node procedure modified. 3. The Install U^xS Personality on a Remote Node procedure modified. 4. Added a new section, "Installing Manuals on a Local Node." 5. Added a new section, "Troubleshooting." 6. Added instructions for obtaining the latest version of the "Ignite-UX" software.

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References

Honeywell Documentation

Publication Title	Publication Number	Binder Title	Binder Number
<i>Universal Station^x System Administration Manual</i>	UX11-403	Universal Station ^x	3095
<i>U^xS Upgrade Installation Instructions</i>	MP-ZRSKUG	Universal Station ^x	3095
<i>Universal Station^x (Ergonomic) Service</i>	UX13-530	Universal Station ^x	3095
<i>Universal Station^x Service</i>	UX13-510	Universal Station ^x	3095
<i>Five/Ten-Slot Module Service</i>	LC13-500	LCN Service - 2	3060-2
<i>Hardware Verification Test System HVTs-Universal Station^x</i>	SW13-511	LCN Service - 3	3060-3
<i>U^xS Specification and Technical Data</i>	UX03-503	System Summary-2	3010-2
<i>System Site Planning</i>	SW02-550	System Site Planning-1	3020-1
<i>LCN Guidelines</i>	LC09-510	LCN Installation	3025
<i>LCN System Installation</i>	SW20-500	LCN Installation	3025
<i>LCN System Checkout</i>	SW20-510	LCN Installation	3025

Hewlett Packard Documentation

Hewlett-Packard CD-ROM Documentation

Hewlett-Packard Documentation at <http://www.docs.hp.com>

1. Universal Station^x Release 300

1.1 Release 300 Differences

The major change to the Release 300 of U^xS is the replacement of the 9.05 version of the HP-UX operating system with the 10.20 version in order to become year 2000 compliant. In addition, the new features provided by the HP-UX upgrade to version 10.20 are listed below.

- A new file system structure, which is modeled after AT&T SVR4 and OSF/1. This filesystem, commonly referred to as V.4 file system, is common to many other UNIX vendors.
- Maximum file size and file system size has been increased to 128 gigabytes.
- The default shell is now the POSIX shell, replacing the Bourne shell that was the default shell in 9.05. Refer to the discussion of shells in Section 5 of the *Release Notes for HP-UX 10.20* for more information.
- The Logical Volume Manager (LVM) is the method of managing disks that are 4 gigabytes or larger. See Section 2.2 of this manual for more information on LVM, and also Chapter 3, “Managing Disks Using the Logical Volume Manager (LVM),” of the *HP-UX System Administration Tasks* manual on the Hewlett-Packard CD-ROM. Please make note of the following LVM drive requirements:
 - LVM is required for root drives that are 4 gigabytes or larger.
 - LVM should not be configured for hard drives that are less than 2 gigabytes.
- Ignite-UX is a new tool that can be used to do initial configuration and installation of HP-UX systems. It replaces the existing HP-UX Install program. Ignite-UX can also be used to aid in disaster recovery. A bootable tape that is the mirror image of your disk can be created and used to restore your system, if necessary.
- Software Distributor is now used for installing and updating software.
- New software installation tools
 - *swinstall* can be used in addition to *update*
 - *swcopy* replaces *updist*
 - *swlist* can be used to list filesets and versions installed on the system.
- Fixes in HP-UX 10.20 to make it Year 2000 compliant.

- Y2K and HP-UX 10.20 compliance for all Honeywell-supplied applications
 - PCDE server and Wonderware NetDDE for UNIX
 - FileLink
 - Multiple Schematics
 - UXS Core software
 - UXS Personality

1.2 Contents of this Release

The following table provides titles and brief descriptions for each of the software DAT tapes that are included in this release.

Tape Title	Brief Description
UXS Install Tape for HP-UX 10.20	Contains a bootable image of HP-UX Install, plus the UXS Ignite image.
UXS Core Tape for HP-UX 10.20	Contains a complete copy of the HP-UX 10.20 CD-ROM, the HP-UX Ignite-UX server, the UXS Core Software, and a UXS Ignite image.
UXS/AXM Patches for HP-UX 10.20	Contains a complete copy of the HP-UX 10.20 recommended patches, plus the Y2K patch bundle.
HP-UX Upgrade Tools	Contains a copy of the HP-UX Upgrade Tools CD-ROM.
Multiple Schematics 120 on HP-UX 10.20 (R4xx)	• UNPx • LCN Daemon • QLT • Boot Stub
Multiple Schematics 300 on HP-UX 10.20 (R5xx, R6xx)	• UNPx • LCN Daemon • QLT • Boot Stub
UXS Personality for R300 HP-UX 10.20 (R6xx)	• UNPx • LCN Daemon • QLT • Boot Stub
UXS Personality for R300 HP-UX 10.20 (R5xx)	• UNPx • LCN Daemon • QLT • Boot Stub
UXS Personality for R300 HP-UX 10.20 (R4xx)	• UNPx • LCN Daemon • QLT • Boot Stub

1.3 General Procedure for Upgrading to Release 300

Step	Action	Applicable Documentation
1	Read Sections 1 and 2 of this manual	<i>U^XS Customer Release Guide</i> Sections 1 and 2
2	Read all relevant HP-UX documentation	HP CD-ROM documentation HP documentation at http://www.docs.hp.com Section 2.1 for a recommended reading list
3	Run the HP Prepare Tool, if necessary.	<i>U^XS Customer Release Guide</i> Section 2.4
4	Gather all preliminary system configuration information.	<i>U^XS Customer Release Guide</i> Section 3.2
5	Perform all necessary backups.	<i>U^XS Customer Release Guide</i> Section 3.4
6	Upgrade a local node to HP-UX 10.20 using a local DAT tape drive.	<i>U^XS Customer Release Guide</i> Section 3.5
7	Install the UXS Personality on a local node using a local DAT tape drive.	<i>U^XS Customer Release Guide</i> Section 3.5
8	Install additional non-operating system software on a local node using a local DAT tape drive.	<i>U^XS Customer Release Guide</i> Section 3.5
9	Restore all customized files on the local node.	<i>U^XS Customer Release Guide</i> Section 3.5
10	Install Ignite UXS Software on the local node that will act as the Ignite-UX server using a local DAT tape drive.	<i>U^XS Customer Release Guide</i> Section 3.6
11	Configure a local node to act as an Ignite-UX Server in order to perform upgrades to HP-UX 10.20 on remote nodes.	<i>U^XS Customer Release Guide</i> Section 3.6
12	Use the Ignite-UX Server to upgrade remote node.	<i>U^XS Customer Release Guide</i> Section 3.6
13	Configure a local Software Distribution Depot in order to upgrade the remote node.	<i>U^XS Customer Release Guide</i> Section 3.6
14	Use the Software Distribution Depot to upgrade non-operating system software on the remote node.	<i>U^XS Customer Release Guide</i> Section 3.6
15	Restore any customized files on the remote node.	<i>U^XS Customer Release Guide</i> Section 3.6
16	Perform any necessary system administration tasks, such as system and data backups.	<i>U^XS System Administration manual</i> Section 7

2. Release 300 Upgrade Preparation

2.1 Important Preliminary Reading

We recommend that anyone who will be using the HP-UX operating system as an operator, developer, or administrator read the appropriate HP documents listed in the following table. It is especially important to read these documents when upgrading your system from HP-UX 9.05 to 10.20 as the migration from 9.05 to 10.20 is not a simple upgrade procedure. These documents are available on the HP CD-ROM or at the web site <http://www.docs.hp.com>.

Table 2—1 Recommended HP-UX Documentation

HP Document	Importance
<i>HP-UX System Administration Tasks</i>	Important as a general reference for system administrators. Section 3 is especially important as a source of information about the Logical Volume Manager.
<i>HP-UX 10.0 File System Layout White Paper</i>	Important as a general reference for file system layout. Chapter 3 contains important information about the system start-up and shutdown model and about adding custom subsystem to this model. If you are upgrading from 9.05 to 10.20, this document is especially important because there are major file system layout changes in 10.20.
<i>Release Notes for HP-UX 10.20</i>	General reference for HP-UX 10.20 functionality. See the discussion of shells (the POSIX shell has replaced the Bourne shell as the default shell in 10.20).
<i>HP-UX 9.x/10.x Interoperability Guide</i>	Important if you will have a mixed system (9.05 and 10.20) for a period of time.
<i>Installing and Updating HP-UX 10.x</i>	This document contains valuable information on updating the HP-UX operating system. Additionally, it contains several chapters detailing the use of an Ignite-UX server.

Table 2—1 Recommended HP-UX Documentation

HP Document	Importance
<i>Upgrading from HP-UX 9.x to 10.x</i>	This document describes the procedure to upgrade from 9.x to 10. 01 Another step would be required to upgrade from 10.01 to 10.20. Honeywell has chosen to do the upgrade by doing a "cold install" of 10.20, eliminating the intermediate step. Nevertheless, this document contains a great deal of useful information and should be read before upgrading from 9.05 to 10.20.
<i>Managing HP-UX Software with SD-UX</i>	This document describes the procedures for managing software installations using HP-UX's software management commands.

2.2 Logical Volume Manager (LVM)

Overview of LVM

LVM is a disk management subsystem that lets you allocate disk space according to the specific or projected sizes of your file systems or raw data. Logical volumes are the preferred method of managing your disks, and are required in certain configurations.

LVM logical volumes provide capabilities that are not possible when using fixed partitions:

- More flexible and efficient ways to organize your disk(s)
- Supports disk striping which can increase I/O performance
- Supports mirroring which can increase data reliability and system availability (In order to use mirroring, you will need to purchase MirrorDisk/UX, HP product number B2491A.

For more information on LVM

If you are planning to use LVM, or need to decide whether or not to use it, see Chapter 3, “Managing Disks Using the Logical Volume Manager (LVM),” of the *HP-UX System Administration Tasks* manual on the Hewlett-Packard CD-ROM or on their documentation web page, <http://www.docs.hp.com>.

Logical Volume Assignments

Table 2—2 Logical Volume Assignment for the 4 GB Drive

Logical Volume	Volume Size Mbytes	Mount Point
/dev/vg00/lvol1	48	/stand
/dev/vg00/lvol2	128	(swap)
/dev/vg00/lvol3	84	/
/dev/vg00/lvol4	32	/tmp
/dev/vg00/lvol5	20	/home
/dev/vg00/lvol6	164	/opt
/dev/vg00/lvol7	52	/users
/dev/vg00/lvol8	332	/usr
/dev/vg00/lvol9	160	/var

This configuration assigns approximately 25% of the disk space, and the remaining space is available for additional volumes or for expansion of existing volumes. In addition, there is considerable unused space available within the existing volumes.

Configuring a Second Hard Drive

If you have a second hard drive, the use of LVM on the second drive is optional. You can convert the second drive to LVM at any time. LVM and non-LVM drives can coexist on your system, but a given physical drive must be either all LVM or all non-LVM. The procedure to convert a non-root disk to LVM is also covered in Chapter 3 of *HP-UX System Administration Tasks*.

2.3 HP-UX 10.20 System Start-up and Shutdown Overview

Start-Up and Shutdown Overview

The method of starting and stopping subsystems is radically different in HP-UX 10.20 than it was in HP-UX 9.05. This section contains a brief overview of the differences, as well as an overview of the new (10.20) model. Developers and administrators who will be writing start-up and shutdown scripts should reference the more comprehensive write-up in Chapter 3 of the *HP-UX 10.0 File System Layout White Paper* for additional information.

HP-UX 9.05 vs. 10.20 Startup

In HP-UX 9.05, initialization and start-up of subsystems was performed by `/etc/rc`— either directly executed from `/etc/rc` or by other scripts called from `/etc/rc`. In HP-UX 10.20, the start-up/shutdown architecture separates the execution scripts from the configuration information required for execution. The file structure used to implement this model has three key elements, as shown in the following table.

Table 2—3 HP-UX Start-up/Shutdown Key Files

File	Function
<code>/sbin/init.d</code>	This directory contains a pool of execution scripts
<code>/etc/rc.config.d</code>	This directory contains configuration files. Each configuration file corresponds to an execution script in the <code>/sbin/init.d</code> directory, and it has the same name as the execution script.
<code>/sbin/rc#.d</code> where # = run level	These are the sequence directories. Each directory corresponds to a run level. These directories contain symbolic links that point to the execution scripts in <code>/sbin/init.d</code> .

All subsystems are started and stopped from a pool of scripts located in `/sbin/init.d`. The scripts get input from configuration files located in `/etc/rc.config.d`. The system run level (set in `/etc/inittab`) determines which subsystems are executed. `/sbin/rc#.d` (e.g., `rc1.d`, `rc2.d`, `rc3.d`) are directories that correspond to the run level of the system. For a system in run level 3, all subsystem in `rc1.d`, `rc2.d`, and `rc3.d` will be executed.

The following diagram depicts the HP-UX 10.20 start-up/shutdown model.

	Run Level Directories /sbin/rc#.d	Pool of Execution Scripts /sbin/init.d	Configuration Files /etc/rc.config.d
rc1.d	->K100phd ->S100localmount	lp cron tdc3000 filelink phd	lp cron tdc3000 filelink phd
rc2.d	->K900nfs.server ->S008net.sd		
rc3.d	->S100nfs.server ->S240hwlicense		
Rc4.d			

The configuration variable files located in `/etc/rc.config.d` are used to set variables to control the execution of the startup script. A control variable is always included to determine whether the subsystem is enabled or disabled. Other variables that may be required to start the subsystem can be included. An example of a configuration file is `/etc/rc.config.d/cron`:

```
#!/sbin/sh
# @(#) $Revision: 72.3 $
# Cron configuration.  See cron(1m)
#
# CRON:          Set to 1 to start cron daemon
#
CRON=1
```

The files in the sequencer directories (`/sbin/rc#.d`) are simply links pointing to the execution scripts. They are separated into run level directories, as well as numbered to control the start order within each directory. Also included in the filenames are **S** (start) or **K** (kill) labels, which determine if the script runs at start-up or shutdown.

The Execution scripts (`/sbin/init.d`) read the corresponding file in `/etc/rc.config.d` to set the variables necessary for execution of the code.

Sequencer Directory

The `/sbin/rc` program is invoked by `init`, and runs the execution scripts when transitioning from one run level to another. `/sbin/rc` goes through all levels between the start and target run level (except run level 0, which is a special case—see the HP white paper). When moving up to a run level, all scripts beginning with `S` will be executed with a start argument for the subsystem. When moving down to a run level all scripts beginning with `K` will be executed.

Example of a sequencer link:

```
/sbin/rc3.d/S241tdc3000
```

The naming convention is:

<code>/sbin/rc3.d/</code>	The link is in the directory <code>/sbin/rc3.d</code> which corresponds to run level 3
<code>S</code>	The sequencing type (<code>S</code> is for start and <code>K</code> is for kill)
<code>241</code>	The sequence number
<code>tdc3000</code>	Reference name of the script that is to be executed (normally, this would be the same as the name of the script, but it does not need to be)

Within a run level directory (`/sbin/rc#.d`), scripts will be executed in order of the sequence number assigned in the link. For example, a script with a sequencer number of 100 will be executed before a script with a sequence number of 200 or 300 or 351, etc.

Execution Scripts

The execution scripts are always invoked with one of four arguments, which indicates the “mode”:

start Bring subsystem up

start_msg Message to print upon startup. This appears in the system startup checklist.

stop Bring subsystem down.

stop_msg Message to print upon shutdown.

The start link **S** and the kill link **K** in `/sbin/rc#.d` point to the same script and the argument is given to the script accordingly.

The execution scripts all have the same format, which allows you to use a template when creating your own startup scripts. `/sbin/init.d/template` is provided for this purpose. The file can be edited to your specific subsystem startup. This template is too long to list here in its entirety. The following is a very condensed version indicating the general structure of the template.

Abridged Listing of the Execution Script Template

Attention

This is an abridged listing of the execution script template. Do not use this abridged version to create a script. Use the full template located in /sbin/init.d/template.

```
case $1 in
'start_msg')
    echo "Starting <specific> subsystem";;
'stop_msg')
    cho "Stopping <specific> subsystem";;
'start')
    if [ -f /etc/rc.config ] ; then
        . /etc/rc.config
    if [ $CONTROL_VARIABLE != 1 ] ; then
        rval=2
    else
        commands for starting subsystem
    fi
    ;;
'stop')
    if [ -f /etc/rc.config ] ; then
        . /etc/rc.config
    if [ $CONTROL_VARIABLE != 1 ] ; then
        rval=2
    else
        commands for stopping subsystem
    fi
    ;;
*)
    echo "usage:  $0 {start|stop}";;
esac
```

When /sbin/rc runs through the startup sequence, it will execute the scripts first with the start_msg argument, then with the start argument.

Sample Procedure Outline for Creating a Start-up/Shutdown Script

The following is an outline of the steps that might be taken to create a startup/shutdown script for an application called `new_product`. To see a completed script, list one of the files in `/sbin/init.d`.

Table 2—4 Example of Creating an Execution Script

Step	Action
1	<code>cd /sbin/init.d</code>
2	<code>cp template new_product</code>
3	<code>vi new_product</code>
4	Edit the <code>start_msg</code> case statement as follows: Replace <code><specific></code> with <code>new_product</code>
5	Edit the <code>stop_msg</code> case statement as follows: Replace <code><specific></code> with <code>new_product</code>
6	Edit <code>start</code> case statement as follows: - Change <code>CONTROL_VARIABLE</code> to <code>NEW_PRODUCT</code> - Add command: <code>/opt/TDC_Open/new_product/bin/new_product.start</code> after the line that states: <i># Execute the commands to start your subsystem</i> (Multiple commands or control commands are allowed) - Add <code>set_return</code> after command/commands
7	Edit the <code>stop</code> case statement as follows: - Change <code>CONTROL_VARIABLE</code> to <code>NEW_PRODUCT</code> - Add command: <code>/opt/TDC_Open/new_product/bin/new_product.stop</code> after the line that states: <i># Execute the commands to start your subsystem</i> (Multiple commands or control commands are allowed) - Add <code>set_return</code> after command/commands
8	<code>vi /etc/rc.config.d/new_product</code>
9	Add single line: <code>NEW_PRODUCT=1</code>
10	Create the links as follows: <code>ln -s /sbin/init.d/new_product /sbin/rc3.d/S900new_product</code> <code>ln -s /sbin/init.d/new_product /sbin/rc2.d/K100new_product</code>

2.4 Running the HP "Prepare" Tool

Overview of HP Prepare

Please note that running the HP "Prepare" Tool is not necessary when upgrading a standard U^XS from Release 200 to Release 300. It is only necessary if there are applications on the U^XS that were installed by the end user. If you have implemented your own applications, or make use of customized scripts, you should, however, perform a preliminary analysis using the HP Prepare tool, which is on the HP-UX Upgrade Tools Tape. This tape is part of the upgrade kit and it is referenced in the backup procedure in the Release 300 Upgrade section of this document.

The HP "prepare" software, which flags possible 10.x incompatibilities, can be run on any 9.05 machine that holds the source code for applications and/or customized scripts. The complete instructions for use of this tool can be found on the Internet at <http://www.docs.hp.com>. This web site is available to everyone and contains HP Technical Documentation. To find this information at this site, select *HP-UX Operating System*, *HP-UX Release 10.x Documentation*, *Upgrading from HP-UX 9.x to 10.x*, Chapter 5, "Converting Code and Scripts."

Checking Text Files for 10.x Incompatibilities

This tool should be run on source code, scripts, and makefiles to check for:

- Invalid pathnames. Pathnames constructed by variable substitution will not be caught (see the *HP-UX 10.0 File System Layout White Paper*).
- Commands, libraries and system calls that have changed (see *Upgrading from HP-UX 9.x to 10.x*, Chapter 8, "Compatibility between 9.x Releases and 10.01"). This tool will not catch changes to commands and libraries that have occurred since 10.01. Consult *Release Notes for HP-UX 10.10* for these. Also, not all networking services are covered, and changes to graphics and X libraries are not covered (see *Release Notes for HP-UX 10.10* for this information).
- Device file names that have changed (see "I/O Convergence" in the *Release Notes for HP-UX 10.10*).

It is not necessary to update references to system files that have changed directory locations. HP-UX 10.x contains transition links from their old location to the new location.

Running the Prepare Tool

Table 2—5 Procedure to Run the HP Prepare Tool

Step	Action
1	Login as engineer on the U ^x S that is still running HP-UX 9.05.
2	On the U ^x S display, select: System Menu > Configuration > System Terminal
3	On the System Terminal window, type the password for root and then press OK.
4	Insert the HP-UX Upgrade Tools tape into the DAT tape drive.
5	Load the upgrade tools by typing the following at the prompt: update
6	Add /upgrade/bin to PATH variable. (setenv PATH /upgrade/bin:\$PATH)
7	- Run <code>prepare</code> once without options by typing the following at the prompt: prepare This creates the <code>.preparerc</code> file that the tool will require later. - Answer <code>Y</code> when asked if you want to create the file.
8	- Review the <code>.preparerc</code> file that you just created (this file is copiously self-documented). - Check the defaults for file types and directories to be included and excluded to ensure that you get desired results

Table 2—5 Procedure to Run the HP Prepare Tool

Step	Action
9	Next, run <code>prepare</code> on a directory specified by <i>directory_name</i>. You can run the tool on all files by specifying <code>/</code> for <i>directory_name</i>. However, it is better to specify a subtree here (for example, <code>/users</code>), to reduce the amount of output the process generates. By specifying an appropriate subtree, you will avoid generating output for system and Honeywell directories. <pre>prepare -bvr directory_name</pre> where: -b tells <code>prepare</code> to do only a quick check of files to see if they need further analysis and conversion. -v tells <code>prepare</code> to report progress to <code>stdout</code>. -r tells <code>prepare</code> to examine <i>directory_name</i> and the directories under it, recursively. This command examines all of the files in your <i>directory_name</i> tree and sorts the files into four categories, creating a file for each category. Each category file contains a list of filenames as follows: 1.SHELL.g this file contains a list of shell scripts and makefiles 1.SRC.g this file contains a list of source files 1.TEXT.g this file contains a list of text files 1.ELSE.g this file contains a list of unrecognized files Each of the files in these lists will have been subjected to a quick check for potential 10.x changes. Four additional category files are created that contain subsets of the above lists. These subsets contain only those files that look as if they will need at least one change. These lists are contained in the following files: 1.SHELL shell scripts and makefiles that potentially require changes 1.SRC source files that potentially require changes 1.TEXT text files that potentially require changes 1.ELSE unrecognized files that potentially require changes
10	Edit each of these lists (1.SHELL, 1.SRC, 1.TEXT and 1.ELSE), removing files you don't want <code>prepare</code> to look at, such as system files. If you used <code>/</code> for the directory name in the previous step, you will see a large number of system files included in the list. These should be deleted.

Table 2—5 Procedure to Run the HP Prepare Tool

Step	Action
11	In this step, you will run the HP Prepare tool on each of the lists of files. NOTE: The prepare tool will duplicate not only your files, but also the original directory tree, under the output directory you specify. If you do not have enough disk space to create copies of the files, omit <code>-m NEWDIR</code> from the commands shown below. Use the following commands to analyze the files in each list, creating new files under NEWDIR. If your AXM is on line, you should issue only one command at a time to keep the load to a minimum. <pre>prepare -V -l+2 -L \$ -S \$ -f 1.SHELL -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.SRC -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.TEXT -m NEWDIR prepare -V -l+2 -L \$ -S \$ -f 1.ELSE -m NEWDIR</pre> where: <code>-V</code> tells prepare to report progress (at the file level) to stdout. <code>-l+2 -L \$</code> creates a logfile 1.SHELL.log <code>-S \$</code> creates a summary file 1.SHELL.sum <code>-f</code> specifies the name of the file that contains the list of files to be examined. <code>-m</code> creates a new version of each original file under NEWDIR. The new version has 10.x HP-UX pathnames in place of 9.x pathnames, and contains comments on other potential problems, such changed options to commands or system calls. <code>-m</code> creates NEWDIR if it does not exist; if it does exist, old files are not overwritten by new ones. (Use <code>-M</code> instead of <code>-m</code> if you want old files to be overwritten.)

Table 2—5 Procedure to Run the HP Prepare Tool

Step	Action
12	Examine the new files (those under <code>NEWDIR</code> in this example) and do the remaining work of determining necessary changes. The tool adds comment lines containing "#analyzer" to the file, in the location where each potential incompatibility is detected. HP includes the tool "fscanner" for locating these comment lines. See the <code>fscanner(update_aid)</code> manpage for more information. You may find "grep" more convenient to use. The appropriate command is: <pre>find NEWDIR -type f -print xargs grep analyzer</pre> The output will show lines like this... <pre>NEWDIR/users/dlb/cdda/alltestc.c:#analyzer, Line 178: Item "exec", WARNING, see LIBS_DB entry L_165</pre> All OpenDDA programs will contain a line for each usage of "exec". This can be ignored. To look up the meaning of an error message, you will need to consult the text files <code>/upgrade/lib/CMDS_DB</code> and <code>/upgrade/lib/LIBS_DB</code>. Search for the string mentioned in the comment line. Using the example above... <pre>pg /upgrade/lib/LIBS_DB</pre> then search for <code>L_165</code>. The HP upgrade tools also provide <code>fnlookup</code>, which maps 9.x pathnames to 10.x pathnames. You can use the command (substitute an absolute pathname below): <pre>fnlookup -o 9.x_absolute_path.</pre> It will return the 10.x pathname.

2.5 HP-UX Patches

Description

The U^xS/A^xM Patches for HP-UX10.20 tape contains a collection of certified HP-UX application patches and recent Y2K upgrade patch files for both an A^xM and a U^xS. However, all necessary patches will automatically be installed when you upgrade to HP-UX 10.20. You may or may not need to install additional optional patches from the tape depending upon the software you have installed on your system. If you do need to install any optional patches, you will use the *swinstall* utility to install the required patches from this tape.

Patch Text Files

Each patch also has an accompanying patch .text file. The following table describes the format of the patch .text files.

Table 2—6 Patch .text Files Format

INFORMATION TYPE	DESCRIPTION
Patch Name:	The name of the patch. Format is: <i>PHxx_yyyy</i> where: <i>PH</i> = Patch HP-UX. <i>xx</i> = area patched: <i>CO</i> - general HP-UX commands. <i>KL</i> - kernel patches. <i>NE</i> - network specific patches. <i>SS</i> - all other subsystems: X11, Starbase, etc. <i>yyyy</i> = a unique number Example: <i>PHSS_4014</i> - an HP-UX subsystem patch name.
Patch Description:	A one-line description that describes this cumulative patch.
Creation Date:	The date that the patch was created.
Post Date:	The date that the patch was posted for general distribution.
Hardware Platforms - OS Releases:	The hardware platforms and HP-UX OS releases on which this patch can be installed.
Products:	This field lists the product name and all product revisions to which this patch applies if it is a patch for an optional product, that is, a non-core operating system product. If the patch is for the core operating system, the value in this field is "N/A".

The patch .text files contain additional types of information not listed in the above table. See *the HP-UX 10.0 Patch Program White Paper* to learn more about patch .text file contents.

Additional Patch Documentation

The following documents, available on the HP-UX documentation CD-ROM and at the <http://docs.hp.com/> website, contain additional information on HP-UX patches:

Location	Document
HP-UX 10.* Release Documentation → HP-UX 10.* Operating System	HP-UX 10.20 Patch List
HP-UX Operating System → White Papers	HP-UX 10.0 Patch Program White Paper

U^xS Installation Instructions

Because of limited available disk space on a U^xS, we recommend that you install patches individually instead of installing the entire patch bundle. Installing the entire patch bundle could break the U^xS system. All patches for HP-UX 10.20 are installed locally using the *swinstall* command that is part of the Software Distributor utility.

There are seven patches within the Extension Software Patch Bundle that must **NOT** be installed. If these patches, which are listed in step 11 of **Error! Reference source not found.**, are installed they may overwrite Honeywell specific code that is part of the X subsystem.

Table 2—7 Installing HP-UX 10.20 Patches on the U ^x S	
Step	Action
1	Back up your system before installing patches.
2	Shutdown the LCN personality.
3	Insert the U ^x S/A ^x M HP-UX Patches tape into the DAT tape drive.
4	Log in as engineer on the U ^x S that you will be installing the patches on.
5	On the U ^x S display, select System Menu > Update Software > SWInstall
6	In the Local Software Install window, type the password for root and click OK.

Table 2—7 Installing HP-UX 10.20 Patches on the U ^X S	
Step	Action
7	A Specify Source (<i>hostname</i>) window appears. Select Local Tape (or Network Directory/CDROM if your tape drive is remote) as the source and click OK. The SD Install - Software Selection (<i>hostname</i>) window appears with the two patch bundles on the U^XS/A^M HP-UX Patches tape listed.
8	Select Options > Change Options and verify that ONLY the following options have been selected: • Create target path if not already there • Mount filesystems in /etc/fstab or /etc/checklist • Reinstall files even if same one already there • Use checksum when checking if file is the same • Autoselect dependencies when marking software • Enforce dependency analysis errors in agent • Enforce script failures • Enforce disk space analysis errors Click OK after making any necessary corrections to the available options to return to the SD Install - Software Selection (<i>hostname</i>) window.
9	Select Actions > Match What Target Has. Notes: This option will determine for each patch whether or not that patch matches a piece of software currently installed on the system. If the software to be patched does not exist on the system, the patch(es) listed in the window will not be selected for installation. Selecting Match What Target Has will also automatically install the <i>HP-UX Core OS Year 2000 Patch Bundle</i>.
10	Select the <i>Extension Software Patch Bundle</i> and then select Actions > Open Item to bring up a list of the individual patches located within this bundle. Note: Software products are listed in alphabetical order.

Table 2—7 Installing HP-UX 10.20 Patches on the U^xS

Step	Action
11	Find the seven patches listed below and “unmark for install”. These patches must not be installed. They should NOT have a “yes” by the patch name. - PHCO_16961 Xcurses library cumulative patch - PHCO_8621 xargs(1) patch to allow '-' as argument - PHSS_11136 HP VUE 3.0 BMS 5.X Cumulative Patch - PHSS_15423 X11R6 Font Server June98 Periodic Patch (already not marked) - PHSS_15431 Xclients June 1998 Cumulative Patch - PHSS_15841 Xserver cumulative patch (already not marked) - PHSS_16120 X/Motif1.2 Runtime Sep98 Periodic Patch If you accidentally selected any of the seven files listed above for installation, deselect them using Ctrl + Click
12	Select Actions > Install Analysis
13	An Install Analysis (hostname) window appears with the following information displayed: - Target: <i>source host name</i> - Status: <i>Analyzing, Ready or Ready with Errors</i> - Products Scheduled: <i>software filesets marked for install</i> Press OK when the Status indicates Ready or Ready with Errors.
14	The analysis logfile will show the following information: 6 of 209 filesets had errors, 56 of 209 will be skipped, 147 of 209 have no errors or warnings. This is an expected result.
15	A Confirmation (hostname) window appears indicating that installation will now begin and that only those products which passed analysis will be installed. Click Yes to continue.
16	An Install Window (hostname) window appears with the following information displayed: - Target: - Status: - Percent Complete: - Kbytes Installed: - Time Left (minutes): The installation of these patches will take approximately three hours. Do NOT abort the installation process until the Status indicates Completed.

Table 2—7 Installing HP-UX 10.20 Patches on the U ^x S	
Step	Action
17	You may look at the Logfile when the status indicates Completed to determine if there were any errors during installation and the nature of the errors. To do so, perform the following: • Click on the Logfile button. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished, click OK. • Click Done from the Install window.
18	Select File > Exit to exit SD Install.
19	Reboot your system by performing the following: • Select System Menu > Configuration > System Terminal. • Enter the password for root in the System Terminal window and click OK. • Type the following at the prompt: <code>/etc/reboot</code>
20	Reload the LCN personality.

3. Release 300 Upgrade Procedure

3.1 Overview of Procedure

This section provides information on and procedures for upgrading from U^xS Release 200 to U^xS Release 300.

Because the upgrade from Release 200 to Release 300 includes completely updating the HP-UX operating system, you will need to gather some preliminary information, and perform a complete system backup before you begin the upgrade procedure. This section specifies the preliminary information you will need and recommends a strategy for backing up your files.

Directly after the Preliminary Information and Backup Strategy sections are a series of procedures for upgrading the software on both a local node and a remote node. This includes instructions for configuring both an Ignite-UX Server and a Software Distribution Depot. See the *Managing Software with SD-UX* section of the *HP-UX 10.* Operating System* manual for additional information on *swinstall*, *swcopy* and *swremove*.

If necessary, refer to Section 1, Universal Station^x Release 300, for additional information on the differences between HP-UX 9.05 and HP-UX 10.20 and for general upgrade procedures. The remaining sections of this manual contain information on and detailed procedures for upgrading the operating system to HP-UX 10.20 and updating the other non-operating system software that is part of this release.

3.2 Before You Begin the Upgrade

Preliminary Information for Upgrade

Later on in the upgrade procedure, you will be prompted for your network information. Before you begin the backup and upgrade procedures, use SAM to determine the following information for all nodes that will be upgraded:

- Time zone and current time and date
- System hostname
- Internet Protocol (IP) address
- Subnetwork mask
- Default Network Gateway
- Gateway hostname
- Gateway network (IP) address
- If using DNS, the domain name server

Additionally, save cron information to the following file:

```
crontab -l > /etc/crontab.save
```

Recommendations for the Upgrade

Honeywell strongly recommends that you shutdown the U^xS and completely restart it and all your applications BEFORE you begin the upgrade. Sometimes changes are made on-line that are lost after a reboot. Make sure your applications can be restarted on HP-UX 9.05. It is important to begin the upgrade from a known state.

3.3 Hardware Upgrade Requirements

DAT Tape Drive

The first Universal Station^x to be upgraded to Release 300 must have a DAT tape drive local to the station. Once the first station is upgraded, it can then be used to upgrade other stations that do not have a DAT tape drive using a remote upgrade procedure.

VT100 Terminal

This upgrade requires a VT100 emulation capable terminal connected to the console port of the WSI 2I/O board in order to upgrade the first Universal Station^x. This terminal can be a dedicated dumb terminal, a personal computer running VT100 terminal emulation software, or a Hewlett Packard workstation.

For example, select `ntcrt24.exe` if using a Windows NT based platform.

See Section 7.14, “System Console Connection” of the *Universal Station^x System Administration* manual for more information on connecting a console to the U^xS.

⇒ **Note:** When installing HP-UX or booting from a recovery tape, the console screen output may be unreadable when using some terminal emulation software from Microsoft, such as HyperTerm. The installation program issues an escape sequence to the terminal in an attempt to identify it. HyperTerm and Terminal.exe do not respond properly to the escape sequence. We had better results using CRT, a terminal emulation software program that is available for trial use via the World Wide Web at <ftp://ftp.vandyke.com/pub/CRT>. The CRT software responds properly to the escape sequence, and produces legible output when the font is set to Courier 10 pt.

HP-UX 10.20 Hard Disk Requirements

The absolute minimum hard disk size requirement when upgrading a Universal Station^x to HP-UX 10.20 is 525 megabytes. If the U^xS is to be used as an Ignite-UX Server, the minimum hard disk requirement for the server station is 1 gigabyte.

3.4 Backup Strategy

Overview of Strategy

This section gives a brief description of the types of file backups you should have before upgrading to HP-UX 10.20 and also provides the step by step instructions for creating these backups. If you have added any files or software to the disk that you would like to restore after the upgrade, you must add them to the backup file list in this procedure or provide for your own backup and restore procedure.

At the end of the upgrade, it is recommended that you do a complete backup of your system and also create a system recovery tape. The procedures for doing so are outlined in the *Universal Station^x System Administration* manual.

Special File Considerations

The .baseUser directory in each user group (engineer, operator, etc.) contains the default files that will be used to initially configure a new user's own directory. Be aware that Honeywell may update these files during a software release upgrade procedure. If you choose to modify any of these files, be sure to create a backup copy. Your custom changes should be merged back in with the new updated file after the software update is complete.

If you do have problems with the upgrade and need to restore HP-UX 9.05 to the hard drive from a local DAT drive, you must have the U^xS Install tape. However, if you are restoring HP-UX from a remote device that has a DAT drive, you will not need the U^xS Install tape. The *Universal Station^x System Administration* manual for Release 200 (UX11-402) contains procedures for restoring your system to HP-UX 9.05 both locally and remotely. See Section 7, "Coprocessor Administration Tasks," for software restoration procedures.

Attention

To increase your chances of creating a good backup tape, we recommend that you use a DAT cleaning cartridge to thoroughly clean your DAT drive before creating backup tapes. We also recommend that you make AT LEAST two backup tapes.

Backup Number One (Full Backup of all Files)

This is a full backup of your primary drive that can be used to recover the system to HP-UX 9.05 in case you encounter difficulties with the upgrade. This backup is absolutely essential if you are upgrading your existing hard drive. The upgrade to HP-UX 10.20 will result in the complete loss of all data on your primary hard drive, and therefore this backup tape will provide the only means of recovering your system should the upgrade fail for any reason. SAM is the preferred method for creating this backup.

Table 3—1 Procedure for Backup Number 1 (full backup)

Step	Action
1	Connect a terminal to the serial port of the U ^X S to be upgraded (this includes a local or a remote upgrade).
2	If the U ^X S is running only Honeywell provided applications, go to Step 4 of this Procedure. Otherwise, continue with the next step.
3	- If you have implemented you own applications, or make use of customized scripts, you should perform a preliminary analysis using the HP “prepare” tool. See the procedure, Running the HP Prepare tool in Section 2.4 of this manual for complete instructions. - If you plan to use the HP Prepare tool, do so now, before proceeding to subsequent steps.
4	- Verify that all the links are in place by typing the following at the prompt: <code>fsck -p</code> - Type <code>y</code> when you see the following prompt: <code>fsck: /dev/dsk/c201d6s0: root file system</code> <code>continue (y/n)?</code> If there are errors, you will see screen output similar to the following: <pre> /dev/dsk/c201d6s0: FREE INODE COUNT WRONG IN SUPERBLK (FIXED) /dev/dsk/c201d6s0: 8 BLK(S) MISSING /dev/dsk/c201d6s0: BAD CYLINDER GROUPS (FIXED) /dev/dsk/c201d6s0: 16829 files, 0 icon, 357933 used, 51237 free (2965 frags, 6034 blocks) ***** FILE SYSTEM WAS MODIFIED ***** ***** REBOOT HP-UX; DO NOT SYNC (USE reboot -n) ***** </pre>
5	Reboot the system if indicated.

Table 3—1 Procedure for Backup Number 1 (full backup)

Step	Action
6	Issue the following commands: <pre>setenv TERM vt100 unsetenv DISPLAY</pre>
7	Invoke SAM.
8	Go to Singer User Mode by selecting Routing Tasks and then tab to System Shutdown. Go to Singer User State and press ENTER.
9	- Press the TAB key to go to <code>Immediate shutdown</code> - Choose OK and press ENTER.
10	Mount the drive by typing the following at the prompt: <pre>mount -a</pre> NOTE: the <code>bdf</code> command will indicate what is mounted.
11	Create backup number 1: This is a full backup of your primary drive. You must be in <code>Single User Mode</code> to create this backup. See “Terminal Option” in Section 7.4, “Backup the Local Coprocessor Hard Drive,” of the <i>U*S System Administration</i> manual for more information on creating this backup.
12	- Verify that the backup tape does not have errors by typing the following at the prompt: <pre>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</pre> - This step may take some time. - If the tape does not have errors, the following messages will appear in the file <code>/tmp/chktape</code>: <pre>: file marker read : checkpoint record read</pre> - Otherwise, you will receive an error message, and you will need to create a new backup tape.
13	Repeat the necessary steps in this procedure to create at least two separate backup tapes (see the Attention box at the beginning of this section).

Backup Number Two (Configuration & Data Files)

This backup of the configuration and data files that can be restored to the original directories after the upgrade. It is much quicker to restore these files from this backup than to recover selected files from your complete backup.

Table 3—2 Procedure for Backup Number 2 (configuration & data)

Step	Action
1	• Create backup number 2 by backing up the files listed in this procedure. NOTE: Only the directories and files listed below should be part of this backup. However, not all of these files may be present on your system. – /.rhosts – /etc/group – /etc/hosts – /etc/hosts.equiv – /etc/crontab.sav – /opt/TDC_Open/netdde/netdde.ini (if NetDDE is loaded) – /etc/opt/TDC_Open/common/filelink.cfg (if File Transfer is loaded) – /etc/nsswitch.conf – /etc/resolv.conf – /usr/lib/x11/app-defaults/MultSchem – /usr/lib/X11/app-defaults/Native.Allow – /usr/lib/X11/app-defaults/NativeLCNWindow – /usr/lib/X11/app-defaults/Session – /usr/lib/X11/app-defaults/Shutdown.Allow – /usr/lib/X11/app-defaults/UsxZoomWindow – /usr/lib/X11/system.Xdefaults – /usr/lib/X11/system.mwmrc – /usr/adm/inetd.sec – /usr/lib/X11/usxserverrc.OnlyOneScreen (for single CRT stations) – /usr/lib/X11/usxserverrc.OverAndUnder (for vertically oriented screens) – /usr/lib/X11/usxserverrc.SideBySide (for horizontally oriented screens) – /usr/lib/X11/usxserverrc – /users – /etc/opt/TDC_Open/common/personality.config – Point.parameter files if PCDE is loaded – Any other user data files

Table 3—2 Procedure for Backup Number 2 (configuration & data)

Step	Action
	• DO NOT backup the following files: – /users/XUS/engineer/.baseUser – /users/XUS/supervisor/.baseUser – /users/XUS/operator/.baseUser – /users/XUS/viewOnly/.baseUser – .mwmrc file under each of your users home directories • For example: /users/XUS/engineer/engineer/.mwmrc /users/XUS/operator/operator/.mwmrc, etc)
2	• Verify that the backup tape does not have errors by typing the following at the prompt: <pre>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</pre> • This step may take some time. • If the tape does not have errors, the following messages will appear in the file /tmp/chktape: <pre>: file marker read : checkpoint record read</pre> • Otherwise, you will receive an error message, and you will need to create a new backup tape.
3	Repeat this procedure to create at least two separate backups (see the Attention box at the beginning of this section).

Backup Number Three (Reference & Analysis Files)

This backup includes files that you will need for reference and analysis, but that CANNOT BE RESTORED TO THEIR ORIGINAL DIRECTORIES.

Table 3—3 Procedure for Backup Number 3 (reference & analysis)

Step	Action
1	Determine how your 9.05 configuration files have been customized: • Use the "diff" utility to compare the original copies of files located in the <code>newconfig</code> directory with the customized files. • Differences in Honeywell files will be taken care of when performing a standard install. • You should be concerned with customizations made at your site to any of the following files: – <code>inetd.conf</code> – <code>inittab</code> – <code>rc</code> – <code>netnfsrc</code> – <code>services</code> • For example, to compare the customized files with the original files, type the following at the prompt: – <code>mkdir/diff</code> – <code>diff /etc/rc /etc/newconfig/rc > diff/diffrc</code> – <code>diff /etc/services /etc/newconfig/services > diff/diffservices</code>
2	Create backup number 3. You can create a tape backup or backup the files to a second drive if you have one. If you choose a tape backup, use a backup method that backs up files by relative and not absolute pathnames. SAM, which uses fbackup (1m) is recommended. This backup should include the following files: • <code>/.cshrc</code> • <code>/.environ</code> • <code>/.exrc</code> • <code>/.login</code> • <code>/.profile</code> • <code>/.session</code> • <code>/diff</code> (directory created in Step 1) • <code>/etc/checklist</code> • <code>/etc/conf/dfile</code> • <code>/etc/inittab</code> • <code>/etc/inetd.conf</code> • <code>/etc/netlinkrc</code> • <code>/etc/netnfsrc</code>

Table 3—3 Procedure for Backup Number 3 (reference & analysis)

Step	Action
	• /etc/rc • /etc/services • /etc/passwd – NOTE: Use this file for reference only in order to add the custom users that were part of the 9.05 system back into the <code>passwd</code> file. – DO NOT modify the default users that were installed when upgrading to 10.20. • .mwmrc files from users home directories
3	• Verify that the backup tape does not have errors by typing the following at the prompt: <code>frecover -x -N -f /dev/rmt/0m -m >& /tmp/chktape</code> • This step may take some time. • If the tape does not have errors, the following messages will appear in the file <code>/tmp/chktape</code>: : file marker read : checkpoint record read • Otherwise, you will receive an error message, and you will need to create a new backup tape.
4	Repeat this procedure to create at least two separate backups (see the Attention box at the beginning of this section).

3.5 Local Node Upgrade Procedure

Overview of Local Procedure

The procedures in this section may be used to upgrade a U^XS node that has a locally connected DAT tape drive. These procedures cover upgrading the operating system to HP-UX 10.20 and upgrading and loading other non-operating system software from a local DAT tape.

This operating system upgrade will reformat the hard disk drive when it installs the newer version of HP-UX. Any customization done by the end user will have to be repeated after the U^XS release 300 software is installed. See also additional information in the *Installing from Media* chapter of the HP documentation at <http://www.docs.hp.com>. To find this information at the web site, select *HP-UX Operating System, HP-UX Release 10.x Documentation, Installing and Updating HP-UX 10.x*, and then select the appropriate chapter.

Local Upgrade Requirements

This upgrade procedure requires the following:

- Local console terminal
- Local DAT tape drive
- UXS Install Tape for HP-UX 10.20
- UXS Personality for R300 HP-UX 10.20 (Rxxx)
- Other optional tapes include
 - UXS Patches for HP-UX 10.20
 - HP-UX Upgrade Tools
 - Multiple Schematics on HP-UX 10.20

Upgrading to HP-UX 10.20 Locally

In this procedure you will upgrade a local node to HP-UX 10.20 by booting from the local DAT tape drive that contains the U^xS Install tape for HP-UX 10.20. This will allow you to install the HP-UX 10.20 operating system onto the local node and perform necessary system configurations.

This procedure is performed through a local console terminal connected to a U^xS that has a local DAT tape drive.

Attention

If you are installing HP-UX 10.20 on a system currently operating under HP-UX 9.05, **DO NOT** perform this procedure unless you have a full backup of your system, especially the root drive. The cold install procedure will completely erase old data. By having a backup, you can revert to your 9.05 system if you encounter any difficulties with the procedure.

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
1	Verify that you have performed all necessary file backups.
2	Connect the console terminal and DAT drive to the U ^x S (see the U ^x S Service Manual).
3	Insert the U ^x S Install tape for HP-UX 10.20 into the DAT tape drive.
4	Log in to the U ^x S as root through the local console terminal and enter the root password.
5	• Verify that you have all of the system configuration settings for the U^xS. • See the “Preliminary Information for Upgrade” subsection in Section 3.2, if necessary.
6	Type the following at the prompt: /etc/reboot
7	Press the ESC key immediately when you see the following message: Press [ESC] to discontinue the boot process

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
8	The following menu appears: <pre> Auto Boot aborted. ==== MAIN MENU ===== Key Operation ----- 1 Boot From a Device 2 Path Configuration 3 Mode Configuration 4 Interactive Testing 5 Firmware Information 6 Hardware Information 7 System Configuration ... 77 Reset the System 88 Change Mode 99 Restart Auto Boot ----- Press Key, then press [Enter/Return] Select Boot From a Device by pressing 1 and then ENTER </pre>
9	• Select the appropriate tape device: – 2 if single disk system – 3 if dual disk system • Press ENTER.
10	A Welcome to Ignite-UX window appears. This Welcome screen is the first of several screens that will guide you through the installation process. • Press the TAB key to highlight items on the screen and then press ENTER to choose or activate the highlighted item. • Use the Up or Down Arrow keys to navigate through the items in the menu.
11	Select the Install HP-UX option and press ENTER.
12	The User Interface and Media Options window appears. • The default for Source Location Options is Media Only Installation • The default for User Interface Options is Guided Installation • Choose these defaults by pressing the TAB key to go to OK and pressing ENTER The system will load the configuration user interface.

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
13	The Install HP-UX wizard: Select an overall system configuration window appears. • The default Configuration is HP-UX B.10.20 Default. • Accept this default by pressing the TAB key to go to NEXT and then pressing ENTER. NOTE: If you are using the Windows terminal emulator, you may see a one-column display that is not readable. If this happens, press Enter to go to the next screen.
14	The Install HP-UX wizard: Select a system environment window appears. • Select the Archive UXS 30 with HP-UX environment and press ENTER.
15	The Install HP-UX wizard: Select a root disk window appears. • Select the disk at the appropriate SCSI address. • Press the TAB key to go to NEXT and then press ENTER.
16	The Install HP-UX wizard: Specify the amount of root swap space window appears. The Install HP-UX wizard determines memory size and calculates a recommended device swap disk size that will be used unless you change it. Honeywell recommends that the root swap space be at least three times the size of the physical memory. If you wish to change the value calculated by the wizard: • Press ENTER to display a list of predefined selections for swap disk size. • Use the Up/Down Arrow key to navigate through the selections. • Go to OK and press ENTER. • To continue to the next menu, press the TAB key to go to NEXT and then press ENTER.
17	The Install HP-UX wizard: Select a file system type window appears. • Make one of the following selections based upon the size of your hard drive: – If the hard disk is less than 2 gigabytes, then select Whole Disk with HFS as the File System Type. – If the hard disk is 4 gigabytes or greater, then select Logical Volume Manager with HFS as the File System Type. – If the hard disk is at least 2 gigabytes, but less than 4 gigabytes, you may choose either option. • To continue, press the TAB key to go to NEXT and then press ENTER.
18	If you selected Logical Volume in the previous step, the Install HP-UX wizard: Specify root volume group disks window appears. • Select the install program default. • Press the TAB key to go to NEXT and then press ENTER.

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
19	The Install HP-UX wizard: Select a language(s) window appears. • Select the default or display additional options by pressing ENTER. • To continue, press the TAB key to go to NEXT and then press ENTER.
20	The Install HP-UX wizard: Select Additional Software window appears. • There is no additional software. • Press the TAB key to go to NEXT and then press ENTER.
21	The Install HP-UX wizard: Pre-install disk information window appears. • Select the install program default. • Press the TAB key to go to NEXT and press ENTER.
22	The Install HP-UX wizard: Pre-install check information window appears. A new window with WARNING message may appear to let you know that all the data in the disk will be destroyed. • DO NOT continue if you have not performed a full backup of your system. See the Backup Strategy section of this manual. • If you performed a full backup of your disk before starting this installation, press the TAB key to go to NEXT and then press ENTER.
23	The Install HP-UX wizard: System Summary window appears. • Select Show Summary and press ENTER to display the General Summary and Hardware Summary tab. • Press the TAB key to go to the selected window/topic and use the Up/Down Arrow key to scroll through additional information. • If you are satisfied with your choices, press the TAB key to go to FINISH and then press ENTER. This phase of the installation will take about 30 to 45 minutes. The system will boot at the completion of this step.
24	The initial installation is complete when you see the following message: <pre> ----- Welcome to HP-UX! Before using your system, you will need to answer a few questions. The first question is whether you plan to use this system on a network. Answer "yes" if you have connected the system to a network and are ready to link with a network. Answer "no" if you: Plan to set up this system as a stand alone (no networking). Want to use the system now as a stand alone and connect to a network later. ----- Are you ready to link this system to a network? Press [y] for yes or [n] for no, then press Return. See the next step for information on the appropriate response.</pre>

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
25	⇒ NOTE: If you wish to perform remote upgrades, you must configure networking at this time as subsequent procedures will use the configuration information you supply in this step. However, if you choose to set up networking at this time and your network is not connected or other nodes are not yet configured, the bootup process will take considerably longer. • Press Y to link this system to a network. • Supply the following information when prompted: – System name and IP address – Root password – Time zone – Any subnets – Subnet mask – DNS server – NIS configuration – Gateway hostname and IP address • If you answer N, you will be prompted for this information only: – System name – Root password – Time zone • Follow the instructions on the screen to configure all necessary information. • Skip configuration of the Font Server if prompted for this information. ⇒ NOTE: It is possible to change or enter the configuration information at a later time using SAM.

Table 3—4 Upgrade a local node to HP-UX 10.20

Step	Action
26	At the completion of this section, a congratulatory message appears. The system will automatically reboot and the following message appears: HP-UX Start-up in progress If you did not configure networking in the previous step, this will take about 2 to 3 minutes. Otherwise, it may take as long as 20 minutes if you do not have a valid network. • A column of functions (tasks) and a corresponding status column provides status indicators such as: – OK – N/A – FAIL. • If you configured networking, the status column should display either OK or N/A. • If you did not configure networking, the following items will indicate FAIL: – Configure LAN interfaces – NCS broker daemons – Start audio server daemon
27	Remove the U ^X S Install tape from the tape drive.

Installing the U^xS Personality Locally

In this procedure you will install the U^xS Personality on a local node using HP-UX's *swinstall* command and the appropriate UX^s Personality for R300 tape. The personality tape you choose depends upon the TPS system software release you are currently using. After installing the U^xS Personality, the U^xS local node will be fully functional.

This procedure is performed through a local console terminal connected to a U^xS that has a local DAT tape drive. You will need to use one of the following tapes:

- UX^s Personality for R300 HP-UX 10.20 (R4xx)
- UX^s Personality for R300 HP-UX 10.20 (R5xx)
- UX^s Personality for R300 HP-UX 10.20 (R6xx)

Table 3—5 Install U^xS Personality on a local node

Step	Action
1	Connect the console terminal and DAT drive to the U ^x S (see the U ^x S Service Manual), if necessary.
2	Insert the appropriate U ^x S Personality for UX ^s R300 tape into the DAT tape drive.
3	Log in to the U ^x S as root through the local console terminal and enter the root password.
4	Type the following at the prompt: <code>/usr/sbin/swinstall</code>
5	The message Starting the terminal version of swinstall... appears with instructions for navigating within the menus using the keyboard. • If you are not familiar with the use of the keyboard for navigation, read the instructions. • Press ENTER to continue.
6	The install software will scan the default source depot type and will display a Note (hostname) window with a message displayed, and with the OK selection highlighted. Press ENTER to continue.

Table 3—5 Install U^XS Personality on a local node

Step	Action
7	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight the software source of Local Tape and press ENTER. • Use the TAB key to go to Source Depot Path and enter the correct path, if necessary. • Press the TAB key to go to OK and then press ENTER.
8	The SD Install - Software Selection (<i>hostname</i>) window appears. The UXS-PERSONALITY software should be listed in the window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
9	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. • Press ENTER to continue.
10	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: • Target • Status • Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.
11	A Confirmation (<i>hostname</i>) window appears providing you with installation information. • Select Yes and press ENTER.

Table 3—5 Install U^xS Personality on a local node

Step	Action
12	An Install Window (hostname) window appears during the installation process with the following information displayed: • Target: <i>hostname</i> • Status: <i>series of status messages</i> • Percent Complete: <i>advances to 100%</i> • Kbytes Installed: • Time Left (minutes): • Loading Software: <i>software product name</i> The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
13	At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: • Press the TAB key to go to Logfile. • Press ENTER to display all the loaded modules. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
14	You will be returned to the Install Window (hostname) window. Press the TAB key to go to DONE and then press ENTER.
15	The SD Install - Software Selection (hostname) window appears. • Press the TAB key to go to the File menu and press ENTER. • Use the down arrow key to go to Exit and press ENTER.
16	You must reboot the node for the U^xS to be functional. • Type the following at the prompt: <i>/etc/reboot</i>
17	When the reboot is complete, the U^xS will be functional.

Table 3—5 Install U^XS Personality on a local node

Step	Action
18	- Use SAM to restore any previous configuration information to the files from Backup Tape Number 2. - In SAM, select “Set additional parameters “ and then select the “Overwrite newer files” option.
19	- Validate the settings of the <code>/var/adm/inetd.sec</code> file for node security. The file that is provided on the update image does not have any security configured. - There is also a template file available that has maximum security configured, including the prevention of remote logins. Be careful if you use this file: <code>/var/adm/inetd.sec.usx</code>
20	- Restore Backup Number 3 to an alternate directory so that you can use the files for reference: - Invoke SAM - In the Restore function, select “Set additional parameters”. - Deselect “Recover files using full pathname”. - Select “Place files in non-root directory”. - Specify a different alternate directory, such as <code>/tmp</code>. - Merge non-Honeywell customizations back into the system. - Restore your additional user accounts. Merge changes to <code>.mwmrc</code> files into the existing accounts.
21	- Invoke SAM. - Select Networking/Communications > Name Service Switch. - Change the search order so that the <code>/etc/hosts</code> file is searched first and DNS is searched second. Note: This is a requirement for X-layer nodes.
22	- If you selected LVM for your disk configuration, then you may increase the size of the logical volumes as there is unused disk space. - You may increase the size of logical volumes if you run out of space within a logical volume while restoring backups or loading software. See Chapter 3, “Managing Disks Using the Logical Volume Manager (LVM),” of the <i>HP-UX System Administration Tasks</i> manual on the Hewlett-Packard CD-ROM. - You may also add logical volumes. If you created large application directories directly under the root volume, then you must create a new logical volume to hold these directories. Use SAM to do this.

Table 3—5 Install U^xS Personality on a local node

Step	Action
23	Use the <code>chmod</code> command to change permissions. <code>cd /</code> • <code>find /var -type f -print xargs chmod 755</code> • <code>find /var -type d -print xargs chmod 755</code> • <code>chmod 777 /var/tmp</code>
24	Use the command line to add the file <code>/etc/shells</code> Include the following lines: • <code>/sbin/sh</code> • <code>/usr/bin/sh</code> • <code>/usr/bin/rsh</code> • <code>/usr/bin/ksh</code> • <code>/usr/bin/rksh</code> • <code>/usr/bin/csh</code> • <code>/usr/bin/keysh</code> • other shell pathnames that are referenced as startup shells in the <code>/etc/passwd</code> file Note: Steps 24 and 25 achieve the same result. You only need to do one of them.
25	Use SAM to modify the users account: Sam > Accounts for Users and Groups > Users > action > Modify>Start-up Program Choose from the following list: • <code>/usr/bin/sh</code> • <code>/usr/bin/rsh</code> • <code>/usr/bin/ksh</code> • <code>/usr/bin/rksh</code> • <code>/usr/bin/csh</code> • <code>/usr/bin/keysh</code>

Table 3—5 Install U^XS Personality on a local node

Step	Action
26	Non Honeywell vendors may provide optional software that is in the "update" format. You must use the update program to install this software. In order to use the update program, you will need to perform the following: • Use vi or another text editor and open the file /usr/local/etc/runUpdate • Make the following change: From: <pre>if ["" ps -ef grep -c update "" -eq 1]; then if ["" ps -ef grep -c updist "" -eq 1]; then</pre> To: <pre>if ["" ps -ef grep -v grep grep -c update `` -eq 0 ]; then     if [ "" ps -ef   grep -v grep   grep -c updist `` -eq 0]; then</pre> After this change, you will able to run update from the Engineers menu: Update Software > Update...
27	Perform this step to prevent the time zone in the System Terminal window from being frozen on the time zone set at installation and to ensure the time is always in synch with the system time zone. Edit the file /etc/profile and change the Timezone section: From: <pre>if [-r /etc/src.sh] then . /etc/src.sh # set the timezone unset SYSTEM_NAME else TZ=MST7MDT # change this for local time. export TZ</pre> To: <pre>if [-r /etc/src.sh] then . /etc/src.sh # set the timezone unset SYSTEM_NAME else . /etc/TIMEZONE</pre> This will keep the system terminal in sync with the rest of the applications.

Table 3—5 Install U^XS Personality on a local node

Step	Action
28	Perform this step to make the time/date work. - Edit the file <code>/users/XUS/engineer/engineer/.mwmrc</code> - Change the "Set Time/Timezone..." section: From: <pre>"Set Time/Timezone..." f.exec "xsu -title 'Set_Time Timezone' - geom +724+400 -e 'hpterm -ls -title Set_Time/Timezone -geom +291+400 -e /sbin/set_parms timezone &' show 'Set_Time Timezone' &</pre> To: <pre>"Set Time/Timezone..." f.exec "xsu -title 'Time/Timezone' -geom +724+400 -e 'hpterm -ls -title Set Time/Timezone -geom +291+400 -e '/sbin/set_parms timezone &'; '/sbin/set_parms date_time &' show 'Set Time/Timezone' &"</pre> Note: This change uses both backquotes (`) and single quotes ('). - Reboot the system after change is complete.

Installing Other Non-Operating System Software Locally

In this procedure you will install additional non-operating system software on a local node using the SWInstall utility that is accessed through a nested menu within the Engineer Workspace Menu.

This procedure is performed through the display console of a U^XS that has a local DAT tape drive.

Table 3—6 Install Multiple Schematics Locally

Step	Action
1	Login as engineer on the U ^X S node.
2	Insert the appropriate software distribution tape into the local DAT tape drive.
3	On the U ^X S display, select System Menu > Update Software > SWInstall Note: If SWInstall does not appear in the menu, then you have overwritten the new version of <code>.mwmrc</code> with the old version. The <code>.mwmrc</code> file under the <code>.baseuser</code> directory was changed to add the SWInstall and SWCopy utilities to the workspace menu.
4	In the Local Software Install window, type the password for root and click OK.
5	A Note (hostname) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
6	A Specify Source (hostname) window appears. Enter the information for the following options: • Source Depot Type . . . – Click to display additional options – Choose Local Tape • Source Host Name . . . <u>hostname of local node</u> • Source Depot Path . . . • Software Filter: <i>none</i> • Click OK.

Table 3—6 Install Multiple Schematics Locally

Step	Action
7	An SD Install - Software Selection (<i>hostname</i>) window appears with a list of products available on the software distribution tape. • Mark the software for installation by clicking on the software product name. Use Ctrl click to select more than one item. Note: Software products are listed in alphabetical order. • Select Actions > Install (analysis)
8	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Click OK to continue.
9	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: • Target: <i>source host name</i> • Status: <i>Analyzing or Ready</i> • Products Scheduled: <i>software products marked for install</i> Click OK when the Status indicates Ready or Ready with Errors.
10	A Confirmation (<i>hostname</i>) window appears indicating that installation will now begin and that only those products that passed analysis will be installed. Click Yes to continue.

Table 3—6 Install Multiple Schematics Locally

Step	Action
11	An Install (<i>hostname</i>) window appears with the following information displayed: Target: Status: Percent Complete: Kbytes Installed: Time Left (minutes): Loading Software: You may choose to look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: • Click on the Logfile button. A Logfile (<i>hostname</i>) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, click OK. • Click Done from the Install Window.
12	Select File > Exit to exit SD Install .
13	Reboot your system to activate MultSchem by performing the following: • Select System Menu > Configuration > System Terminal • Enter the password for root in the System Terminal Window and click OK • Type the following at the prompt: <code>/etc/reboot</code>
14	Remove the software distribution tape from the local tape drive.

3.6 Remote Node Upgrade Procedure

Overview of Remote Procedure

Upgrading to HP-UX 10.20 on additional U^XS nodes can be done from an upgrade server. This section contains the procedures for configuring the upgrade server in order to perform remote node upgrades. It also contains the procedures for configuring a Software Distribution Depot, which can be used to perform upgrades of non-operating system software on remote nodes.

See also additional information in Ignite-UX Server chapters of the HP documentation at <http://www.docs.hp.com>. To find this information at this site, select *HP-UX Operating System*, *HP-UX Release 10.x Documentation*, *Installing and Updating HP-UX 10.x*, and then select the appropriate chapter. For online information about the Ignite-UX server, please see the `/opt/ignite/share/doc/` directory on your system, and the manpage *ignite(5)*.

The procedures in this section include the following:

- Installing U^XS Ignite Software on a U^XS station to act as a server
- Configuring the U^XS station to act as an Ignite-UX Server
- Installing HP-UX 10.20 to a remote node using the Ignite-UX Server
- Loading software onto a Software Distribution Depot
- Installing non-operating system software to a remote node

Remote Upgrade Requirements

- Ignite UX Server Node must have a hard drive of 1 gigabyte or larger
- Ignite UX Server Node must already have been upgraded to HP-UX 10.20
- Host name and individual IP address of each node to be upgraded
- Host name and individual IP Address of the node that will act as the Ignite U^XS Server

Installing Ignite-UX on a Local Node

In this procedure you will install the Ignite-UX Server software, which is on the U^XS Core tape, onto a local node using the SWInstall utility. SWInstall can be accessed through a nested menu within the Engineer Workspace Menu. After installing the necessary software and configuring the local U^XS node, it can then be used as an Ignite-UX Server in order to upgrade remote nodes.

This procedure is performed through the display console at a U^XS that has a local DAT tape drive and a hard drive with at least 150 MB of available disk space.

⇒ **NOTE:** Loading the U^XS Install tape will automatically install Ignite-UX.

Table 3—7 Install U^XS -Ignite Software on a Local Node

Step	Action
1	Obtain the latest version of the “Ignite-UX” software from the HP web site: • Go to http://www.software.hp.com • Type Ignite in the Search the Catalog text box and then click GO. • Click Ignite-UX in the Product Name/Description column. • Click Download Ignite-UX • Click Download installation instructions • Click Download Ignite-UX for 10.x Version A.2.2 • After download is complete, click FIX .tar.gz • Follow the download installation instructions to complete the install. Note: As of November 1999, the latest revision is A2.2 (10-28-99).
2	Login as engineer on the U ^X S node that will be used as the Ignite-UX Server.
3	Insert the U ^X S Core tape into the local DAT tape drive.
4	On the UXS display, select System Menu > Update Software > SWInstall
5	In the Local Software Install window, type the password for root and click OK.
6	A Note (hostname) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
7	A Specify Source (hostname) window appears. Enter the information for the following options: • Source Depot Type . . . – Click to display additional options – Choose Local Tape • Source Host Name . . . <i>hostname of local node</i> _ • Source Depot Path . . . • Software Filter: none • Click OK.

Table 3—7 Install U^xS -Ignite Software on a Local Node

Step	Action
8	An SD Install - Software Selection (<i>hostname</i>) window appears with a list of products available on the UXS Core tape. - Mark the following software for installation by clicking on the software product name. Use Ctrl click to select more than one item. - UXS-IGNITE Note: Software products are listed in alphabetical order. - Select Actions > Install (analysis)
9	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Click OK to continue
10	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: - Target: <i>source host name</i> - Status: <i>Analyzing or Ready</i> - Products Scheduled: <i>software products marked for install</i> Click OK when the Status indicates Ready or Ready with Errors.
11	A Confirmation (<i>hostname</i>) window appears indicating that installation will now begin and that only those products that passed analysis will be installed. Click Yes to continue.

Table 3—7 Install U^XS -Ignite Software on a Local Node

Step	Action
12	An Install (<i>hostname</i>) window appears with the following information displayed: – Target: – Status: – Percent Complete: – Kbytes Installed: – Time Left (minutes): – Loading Software: You may choose to look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: • Click on the Logfile button. A Logfile (<i>hostname</i>) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, click OK. • Click Done from the Install Window.
13	Select File > Exit to exit SD Install .
14	Remove the U ^X S Core tape from the local tape drive.

Configuring the Ignite-UX Server

In this procedure you will configure the local node on which you previously installed Ignite-UX Server software. Configuring the node will complete the steps necessary to use the node as an Ignite-UX Server in order to upgrade remote nodes.

This procedure is performed through the System Terminal window at the display console of a U^xS that has the Ignite-UX Server software installed on it.

Table 3—8 Configure an Ignite-UX Server

Step	Action
1	On the U ^x S display, select System Menu > Configuration > System Terminal
2	In the System Terminal window enter the password for root and press OK.
3	You will now need to edit the <code>UxS_300_archive_cfg</code> file in order to set the <code>nfs_source</code> to the IP address of the node acting as the Ignite-UX Server. To do so, perform the following from the System Terminal window: - Type the following command: <pre>vi /var/opt/ignite/data/Rel_B.10.20/UxS_300_archive_cfg</pre> Note: The U and S are uppercase; x is lowercase - Locate the following line identifying the <code>nfs_source</code>: <pre>nfs_source = "IP_Address:/var/opt/ignite/archives/UxS.300.0"</pre> - Change the <code>IP_Address</code> to the IP address of the node that will act as the Ignite-UX Server. - Save the modified file and exit the editor.
4	From the System Terminal window, type the following command: <pre>/usr/sbin/sam</pre> The System Administration Manager menu appears.
5	In the SAM window: - Click on the Networking and Communications icon and select Actions > Open Item - Click on the Networked File Systems icon and select Actions > Open Item - Click on the Exported Local File Systems icon and selection Actions > Run. - Select Yes when you receive the following confirmation message: “NFS Server not enabled. Do you wish to enable?”
6	From the Networked File System (hostname) window, select Action > Add Exported File System

Table 3—8 Configure an Ignite-UX Server

Step	Action
7	The Add Exported File System (<i>hostname</i>) window appears. You will need to add the local directories to the Exported Local File Systems list by entering the information specified in the following 3 steps.
8	In the Add Exported File System window, make the following selections: • Local directory name: <code>/var/opt/ignite/clients</code> • Choose Option for Specify UID for unknown User: – Specify UID for unknown – User ID: 2 • Accept the defaults for other options. • Click Apply
9	In the Add Exported File System window, make the following selections: • Local directory name: <code>/var/opt/ignite/data/Rel_B.10.20</code> • Click Specify User Access button • In the User Access (<i>hostname</i>) window make the following selections: – Allow Access From: All Systems – Access Types: Read-only – Click OK • Click Apply
10	In the Add Exported File System window, make the following selections: • Local directory name: <code>/var/opt/ignite/archives/UxS.300.0</code> • Click Specify User Access button • In the User Access (<i>hostname</i>) window make the following selections: – Access Types: Read-only – Click OK • Click Apply
11	Click Cancel to return to the Add Exported File System (<i>hostname</i>) window.
12	Select File > Exit to return to the SAM window.
13	Select File > Exit to exit SAM .

Upgrading to HP-UX 10.20 Remotely Using an Ignite-UX Server

More detailed information on installing from an Ignite-UX Server is contained in Chapter 5 of the Installing and Updating HP-UX 10.x HP Documentation. Use this procedure when you need to upgrade a U^xS node using the Ignite-UX server that has Ignite-UX and U^xS-Ignite installed.

In this procedure you will upgrade a remote node to HP-UX 10.20 using the Ignite-UX Server. This will allow you to install the HP-UX 10.20 operating system onto remote nodes.

This procedure is performed through the System Terminal window at the display console of a U^xS that has the Ignite-UX and UXS-Ignite software installed on it. It is also assumed that the node to be upgraded is currently on line and running HP-UX. Before beginning this procedure, verify that you have all the necessary preliminary information for both the client and the host.

This upgrade procedure will reformat the hard disk drive and install U^xS release 300 on the node to be upgraded. Any customization done by the end user will have to be repeated after the U^xS release 300 software is installed.

Table 3—9 Upgrade Remote Node to HP-UX 10.20

Step	Action
1	Log in as engineer to the U ^x S node that is acting as the Ignite-UX server.
2	On the U ^x S display, select System Menu > Configuration > System Terminal
3	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
4	In the System Terminal window, type the following command: <code>/opt/ignite/bin/bootsys -w -v System_Name</code> (where <code>System_Name</code> is the name of the remote node to be upgraded)
5	Select Y and press ENTER when the following prompt appears: "Would you like bootsys to set the .rhosts file on that system at this time? ([y]/n)"
6	Enter the root password of the remote node to be upgraded. You will receive a series of screen messages.
7	In the System Terminal window, type the following command: <code>/opt/ignite/bin/ignite</code>

Table 3—9 Upgrade Remote Node to HP-UX 10.20

Step	Action
8	If a window appears with the message No clients found , click OK. A Welcome to Ignite UX screen appears. Click OK.
9	An Ignite UX (hostname) window appears. The window displays a series of node icons. It may take a few minutes for the icons to appear. Click on the icon that represents the remote node that is to be upgraded.
10	- Select Actions > Install Client > New Install An opt/ignite/bin/itool window appears with a System Hardware Inventory screen. - Click OK. The opt/ignite/bin/itool screen appears with a series of tabs across the top.
11	Select the Basic tab.
12	Make the following selections from within the Basic tab: - Configurations: UxS 300 with HP-UX 10.20 - Environments: ARCHIVE UxS 300 w/ HP-UX - Root Disk setting and File System Type: - If the disk is 2 GB or less, set the File System Type to Whole Disk with HFS. - If the disk is 4 GB or greater, set the File System Type to Logical Volume Manager with HFS. - If the disk is at least 2 GB, but less than 4 GB you may choose the Whole Disk or Logical Volume Manager option. - Validate that the Root Swap size is set to at least three times the size of physical memory.
13	Select the System tab.
14	Under Final System Parameters select Set Parameters Now and fill in the information for all the fields.
15	Set the following configurations in the System tab: - Time Zone - Root Password - Network Services
16	Select the DNS tab.
17	- Verify that the gateway and DNS information is properly set. - Click Go!

Table 3—9 Upgrade Remote Node to HP-UX 10.20

Step	Action
18	An itool Confirmation (<i>hostname</i>) window appears with a warning message stating that "Continuing the installation will destroy any existing data on this disk." • If the only warning is that the disk appears to contain HP-UX – Press Go! to continue. – Otherwise, correct the problem before continuing.
19	• In the Ignite-UX window, select Action > Client Status to bring up a Client Status window where you may monitor the upgrade process. • The left column of each row will display COMPLETE when the upgrade is finished. Click OK when the upgrade is complete. Note: The upgrade will take approximately 20 minutes if there are no slow gateways.
20	The node may contain some warnings as a result of the upgrade. These warnings are OK.
21	From the Ignite-UX window select File > Exit.

Copying Non-Operating System Software to a Software Distribution Depot

In this procedure you will copy the UXS Personality software to a local node using the SWCopy utility that is accessed through a nested menu within the Engineer Workspace Menu. The `swcopy` command copies the software from the source (local tape in this procedure) to a depot which can then be used as the installation source for remote nodes on the same network. Software that is copied into a depot cannot be used; it is placed there only to act as a source for other installations.

This procedure is performed through the display console of a U^XS that has a local DAT tape drive.

⇒ **Note:** You may use this procedure to copy other non-operating system software to a Software Distribution Depot by substituting the appropriate software product tape, target depot path, and software product name in place of the U^XS Personality references.

Table 3—10 Copy U^XS Personality to a Software Distribution Depot

Step	Action
1	Log in as engineer to the UxS node that will act as the Software Distribution Depot.
2	Insert the appropriate UXS Personality tape into the local DAT tape drive.
3	On the UXS display, select System Menu > Update Software > SWCopy
4	In the Create Software Depot window, enter the password for root and click OK.
5	A Select Target Depot Path (<i>hostname</i>) window appears. Enter the following information: • Target Depot Path: <code>/var/spool/sw/uxs-pers</code> • Click OK
6	Click OK if a note window appears with the following message: "The default source cannot be used as a source because it is already marked as a target."

Table 3—10 Copy U^xS Personality to a Software Distribution Depot

Step	Action
7	• A Specify Source (<i>hostname</i>) window appears. Enter the following information: – Source Depot Type: Local Tape – Source Host Name . . . <i>hostname</i> – Source Depot Path: <code>/dev/rmt/0m</code> – Software filter: none • Click OK
8	In the SD Copy - Software Selection (<i>hostname</i>) window, perform the following: • Select Options > Change Options. • A Note (<i>hostname</i>) window appears that recommends using caution when changing options. Click OK to continue. • Uncheck the Enforce dependency analysis errors in agent. • Click OK.
9	In the SD Copy - Software Selection (<i>hostname</i>) window perform the following: • Highlight the appropriate software selection line. • Select Actions > Copy (Analysis)
10	An Error (<i>hostname</i>) window may appear indicating an error because pre-requisite filesets are required. These filesets are already installed on the target U^xS nodes. Press ENTER to continue
11	Click OK when the Copy Analysis (<i>hostname</i>) window shows a status of <code>Ready with Warnings</code>
12	In the Confirmation (<i>hostname</i>) window, click Yes.
13	A Copy Window (<i>hostname</i>) window appears indicating the following information: – Target: – Status: – Percent Complete: – Kbytes copied: – Time left (minutes): – Copying software: Click Done when the Status indicates completed.

Table 3—10 Copy U^xS Personality to a Software Distribution Depot

Step	Action
14	Exit SWCopy by selecting File > Exit
15	Remove the U ^x S Personality for UxS R300 tape from the local tape drive.
16	The software is now loaded on the Software Distribution Depot. When you are ready to load the software onto a remote node, go to the next subsection, “Loading the UXS Personality from a Software Distribution Depot to a Remote Node.”

Installing Non-Operating System Software to a Remote Node

In this procedure you will install the U^xS Personality on a remote node using HP-UX's *swinstall* command and the Software Distribution Depot configured in the previous procedure. After installing the U^xS Personality, the U^xS remote node will be fully functional.

This procedure is performed through the System Terminal window at the display console of a U^xS that has been configured as the Software Distribution Depot. You will remote login to the client node and then install the software from the Software Distribution Depot. It is assumed that the node to which the software will be installed is on line and has already been upgraded to HP-UX 10.20.

Before beginning this procedure, verify that you have all the necessary preliminary information for both the client and the host.

⇒ **Note:** You may use this procedure to install other non-operating system software to a remote node by substituting the appropriate software product tape, target depot path, and software product name in place of the UXS Personality references.

Table 3—11 Install U^xS Personality on a Remote Node

Step	Action
1	Log in as engineer to the U ^x S node that contains the UXS Personality software loaded on the Software Distribution Depot
2	On the U ^x S display, select System Menu > Configuration > System Terminal
3	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
4	Type the following at the prompt: <code>rlogin client_node_name</code> Where <i>client_node_name</i> is the actual name of the node that will have the software installed.
5	Enter the password of the client node and press ENTER.
6	In the remote System Terminal window, enter the following command: <code>/usr/sbin/swinstall/</code>
7	A screen of instructions explaining how to use <i>swinstall</i> through the terminal appears. Press ENTER to continue.

Table 3—11 Install U^XS Personality on a Remote Node

Step	Action
8	The SD Install - Software Selection (<i>hostname</i>) window appears within the System Terminal Window with a note stating that the software source does not exist. You will define the software source in the next step. Press ENTER to continue.
9	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight the software source of Network Directory and press ENTER. • Use the TAB key to go to Source Depot Path. Enter the Target Depot Path defined in the previous procedure. • Press the TAB key to go to OK and then press ENTER.
10	You will return to the SD Install - Software Selection (<i>hostname</i>) window. The UXS- PERSONALITY software should be listed in the window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
11	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press ENTER to continue
12	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: • Target • Status • Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready . • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.

Table 3—11 Install U^xS Personality on a Remote Node

Step	Action
13	A Confirmation (hostname) window appears with the installation information displayed. Press ENTER to continue.
14	An Install Window (hostname) window appears during the installation process with the following information displayed: • Target: <i>hostname</i> • Status: <i>series of status messages</i> • Percent Complete: <i>advances to 100%</i> • Kbytes Installed: • Time Left (minutes): • Loading Software: <i>software product name</i> The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready .
15	At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: • Press the TAB key to go to Logfile. • Press ENTER to display all the loaded modules. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
16	You will be returned to the Install Window (hostname) window. Press the TAB key to go to DONE and then press ENTER.
17	The SD Install - Software Selection (hostname) window appears. • Use the TAB key to go to the File Menu and press ENTER. • Use the down arrow key to go to Exit and press ENTER.
18	You must reboot the node. Type the following at the prompt: <code>/etc/reboot</code>
19	When the reboot is complete, the remote U ^x S will be functional.
20	Go to Table 3-5, "Install UXS Personality on a Local Node", and perform steps 18 through 28.

3.7 Installing Manuals on a local node

Overview of Procedure

In this procedure you will install the Manuals software, which is on the UxS Core tape, onto a local node using the SWInstall utility. SWInstall can be accessed through a nested menu within the Engineer Workspace Menu. After installing the necessary software, it can then be used.

Perform this procedure from the display console of a U^xS that has

- a local DAT tape drive, and
- a hard drive with at least 7876 Kbytes of available disk space.

Table 3—12 Install manuals on a local node

Step	Action
1	Login as engineer on the U ^x S node.
2	Insert the U ^x S Core tape into the local DAT tape drive.
3	From the U ^x S display, select System Menu > Update Software > SWInstall
4	From the Local Software Install window, type the password for root and click OK.
5	A Note (hostname) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
6	A Specify Source (hostname) window appears. • Enter the information for the following options: – Source Depot Type . . . – Click to display additional options – Choose Local Tape – Source Host Name . . .hostname of local node – Source Depot Path . . . – Software Filter: none • Click OK.

Table 3—12 Install manuals on a local node

Step	Action
7	An SD Install - Software Selection (hostname) window appears with a list of the products available on the U^XS Core tape. - Mark the following software for installation by clicking on the software product name. --B3782EA Note: Software products are listed in alphabetical order. - Select Actions > Open Item
8	- Select each of the manual files one by one. - After making each selection, perform the following - Select Actions > Open Item. - Mark Manuals for installation: - Select Actions > Mark For Install - Select Actions > Close Level Note: You will get a note on the screen” In addition to the software you just marked, other software was automatically marked to resolve dependencies. This message will not appear again.” This is OK. - B3782EA.CDE.Manuals - B3782EA.DCE-Core.Manuals - B3782EA.ImagingSubsystem.Manuals - B3782EA.InternetSrvcs.Manuals - B3782EA.LVM.Manuals - B3782EA.MailUtilities.Manuals - B3782EA.NFS.Manuals - B3782EA.Networking.Manuals - B3782EA.OS-Core.Manuals - B3782EA.PrinterMgmt.Manuals - B3782EA.ProgSupport.Manuals - B3782EA.SW-DIST.Manuals - B3782EA.Streams.Manuals - B3782EA.Streams-TIO.Manuals - B3782EA.SystemAdmin.Manuals - B3782EA.TextEditors.Manuals - B3782EA.TextFormatters.Manuals - B3782EA.UUCP.Manuals - B3782EA.Upgrade.Manuals - B3782EA.X11.Manuals

Table 3—12 Install manuals on a local node

Step	Action
9	Select Action > Install
10	An Install Analysis (hostname) window appears with the following information displayed: Target: <i>source host name</i> Status: <i>Ready</i> Products Scheduled: <i>20 of 20</i> Click OK when the Status indicates Ready. Note: Following is a summary of the analysis phase: “4 of 39 filesets will be Skipped. 35 of 39 filesets had no Errors or Warnings. The Analysis Phase succeeded.”
11	A Confirmation (hostname) window appears indicating that installation will now begin and that only those products that passed analysis will be installed. Click Yes to continue.
12	An Install (hostname) window appears with the following information displayed: Target: Status: Percent Complete: Kbytes Installed: Time Left (minutes): Loading Software: You may choose to look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: • Click on the Logfile button. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, click OK. • Click Done from the Install Window.
13	Select File > Exit to exit SD Install.
14	Remove the U ^X S Core tape from the local tape drive.

3.8 Troubleshooting

Overview

Installation of the U^xS upgrade must be performed on the final target machine. If the upgrade for a dual-screen U^xS is performed on a U^xS with a single screen you must perform this procedure.

It is not necessary, however, to have a dual screen U^xS in order to add the graph3 driver to the kernel and to rebuild the kernel. A single screen U^xS without the graphic is sufficient.

Table 3—13 Add the graph3 driver

Step	Action
1	Once the "Dual Screen" U ^x S configured software is restarted on a dual screen U ^x S, select System Menu > Configuration > System Configuration from the workspace menu.
2	Type the coprocessor (root) password and Return at the prompt. The System Administration Manager (SAM) window appears.
3	Select Kernel Configuration > Drivers
4	From the menu, select graph3
5	Select Actions > Add Driver to Kernel The Pending State changes from Out to In.
6	Select Actions > Create a New Kernel
7	- Check the following: - Move Kernel Into Place and Shutdown/Reboot System Now - Click OK.
8	A Note (hostname) appears. Click OK.

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